



Service Bulletin Number: 5579934

Released Date: 22-jul-2019

Exhaust Gas Recirculation (EGR) System Intake Air Connection Inspection and Cleaning Guidelines

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I. Products Affected

All midrange engines with exhaust gas recirculation (EGR)

II. Introduction

- This document provides guidelines to inspect and clean the intake air connection in the exhaust gas recirculation (EGR) system.
- The intake air connection combines EGR flow and fresh air flow into charge flow. Charge flow then goes through the air intake grid heater prior to reaching the air intake manifold.
- EGR flows through the intake air connection exhaust gas entrance ports prior to entering the fresh air flow stream. A differential pressure is measured across one of the exhaust gas entrance ports. The engine control module (ECM) uses this measurement to estimate EGR flow.
- EGR flow measurement by the ECM is accurate when the intake air connection exhaust gas entrance ports are clean or lightly sooted.
- EGR differential pressure sensor ports/cross drillings need to be maintained in a clean condition to ensure that EGR flow measurement by ECM is accurate.
- Air Intake Connection exhaust gas entrance ports and EGR differential pressure sensor ports/cross drillings may accumulate moderate soot deposits that can change the ECM's EGR flow measurement accuracy.

III. Inspection and Cleaning of Intake Air Connection Exhaust Gas Entrance Ports

- Figures 3, 4, and 5 below show multiple examples of soot accumulation in the exhaust gas entrance ports in various Air Intake Connections.
- Verify the engine is off and that the EGR system has been given time to cool down.
- Remove the Air Intake Connection from the engine. See corresponding Service Manual. Reference Procedure 010-080 in Section 10.

- Inspect the exhaust gas entrance ports on the Air Intake Connection for any soot deposits using Figures 3, 4 and 5 below. Figure 2 below is a reference to how a normal Air Intake Connection should look.
- If moderate soot deposits are observed inside the Air Intake Connection exhaust gas entrance ports proceed to thoroughly clean the Air Intake Connection. Use a nylon brush to clean off any loose soot and use a solvent to clean out the rest of the deposits. Verify that EGR differential pressure sensor ports (vertical and horizontal) are also cleaned out. Reference Section IV in this document. See corresponding Service Manual. Reference Procedure 010-080 in Section 10.
- If shellac (hard, shiny that is not able to be cleaned with a nylon brush) is found on the part proceed to replace the Air Intake Connection.
- After cleaning, install the Air Intake Connection onto the engine. Reference Figure 6 below for a visual on how the Air Intake Connection should look like after proper cleaning. See corresponding Service Manual. Reference Procedure 010-080 in Section 10.
- After the Air Intake Connection has been installed on the engine check for leaks on the intake and the exhaust systems. See corresponding Service Manual. Reference Procedure 010-024 in Section 10.

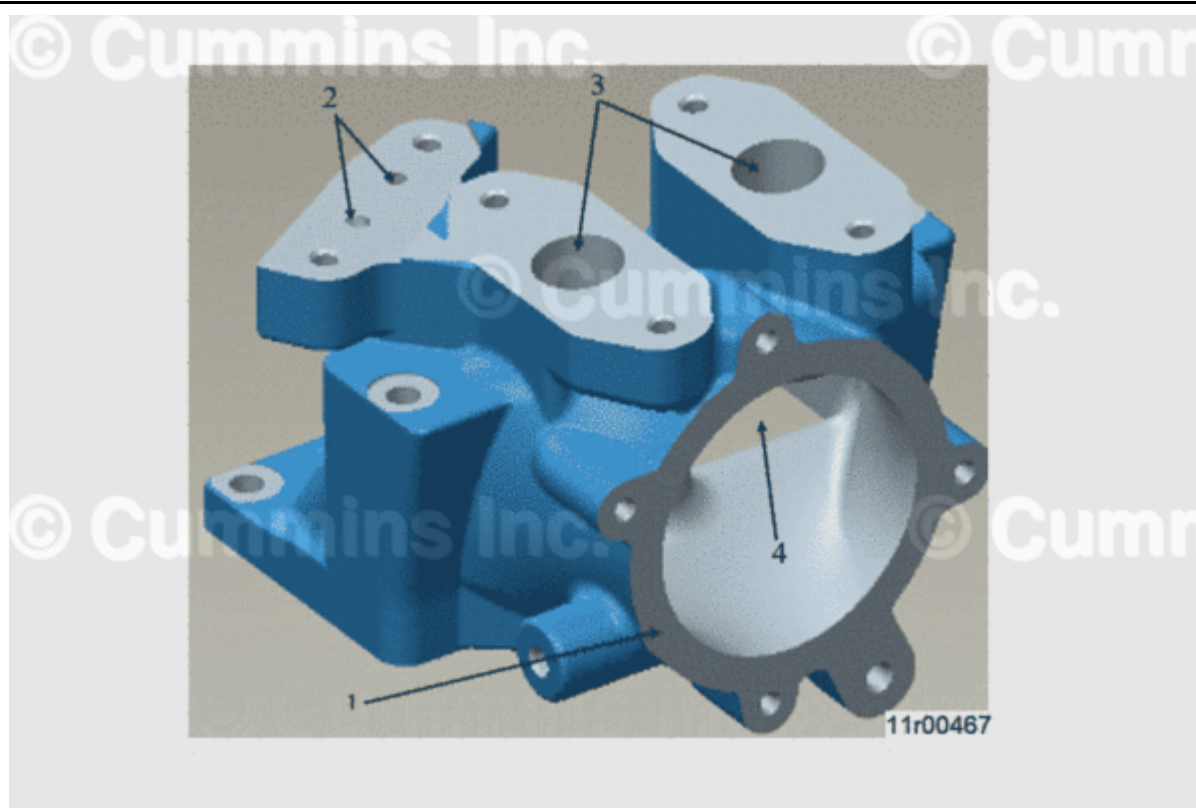


Figure 1, Intake Air Connection Overview.

Table 1, Air Intake Connection Overview

Figure 1, Item Number	Description
1	Intake Air Throttle Connecting Location
2	EGR Differential Pressure Sensor Ports
3	Intake Air Connection Exhaust Gas Entrance Ports

Table 1, Air Intake Connection Overview	
4	Intake Air Grid Heater Connection

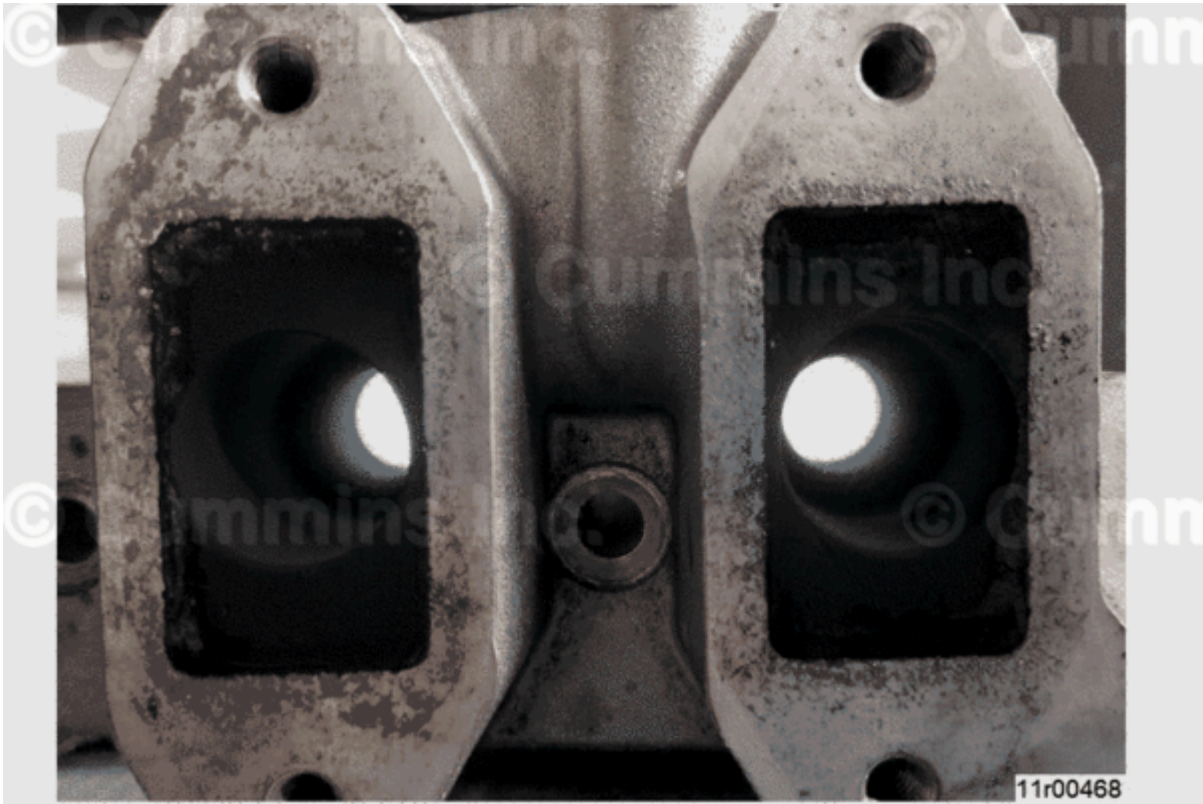


Figure 2, Flow Orifice of a Normal Intake Air Connection.

Table 2, Flow Orifice of a Normal Intake Air Connection	
Appearance	Smooth soot layer inside exhaust gas entrance ports; exit diameter of intake air connection exhaust gas entrance port do not have any obvious/apparent soot accumulation and look smooth and symmetrical.
Cause	Normal engine operation, normal functioning intake air and exhaust systems.
Action	No action, continue to use intake air connection.



Figure 3, Flow Orifice of a Field Return Intake Air Connection With Excessive Soot.

Table 3, Flow Orifice of a Field Return Intake Air Connection

Appearance	Signs of uneven deposits can be seen on the walls of the intake exhaust gas entrance ports; asymmetrical exit diameter is observed to be smaller compared to a normal functioning part.
Cause	Soot accumulation taking place inside intake air connection exhaust gas entrance ports.
Action	Clean intake air connection exhaust gas entrance ports using a nylon brush and solvent.

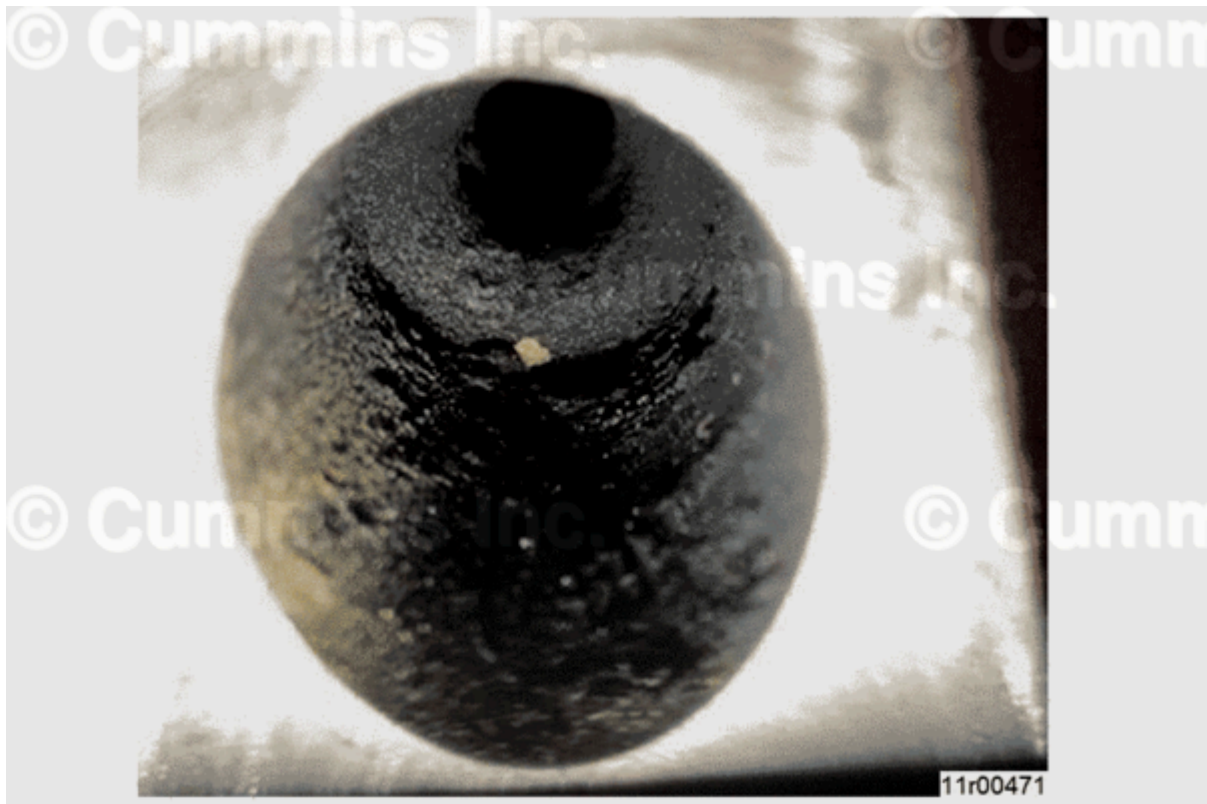


Figure 4, Flow Orifice of a Field Return Intake Air Connection With Excessive Soot.

Table 4, Flow Orifice of a Field Return Intake Air Connection

Appearance	Signs of soot layer unevenness observed with localized soot lumps; exhaust gas entrance port exit diameter is observed to be smoother but smaller when compared to normal intake air connection.
Cause	Soot accumulation taking place inside intake air connection exhaust gas entrance ports.
Action	Clean intake air connection exhaust gas entrance ports using a nylon brush and solvent.



Figure 5, Flow Orifice of a Field Return Intake Air Connection With Excessive Soot.

Table 5, Flow Orifice of a Field Return Intake Air Connection

Appearance	Signs of soot layer unevenness observed with localized soot lumps; exhaust gas entrance port exit diameter is observed to not be smoother but smaller when compared to normal intake air connection.
Cause	Soot accumulation taking place inside intake air connection exhaust gas entrance ports.
Action	Clean intake air connection exhaust gas entrance ports using a nylon brush and solvent.

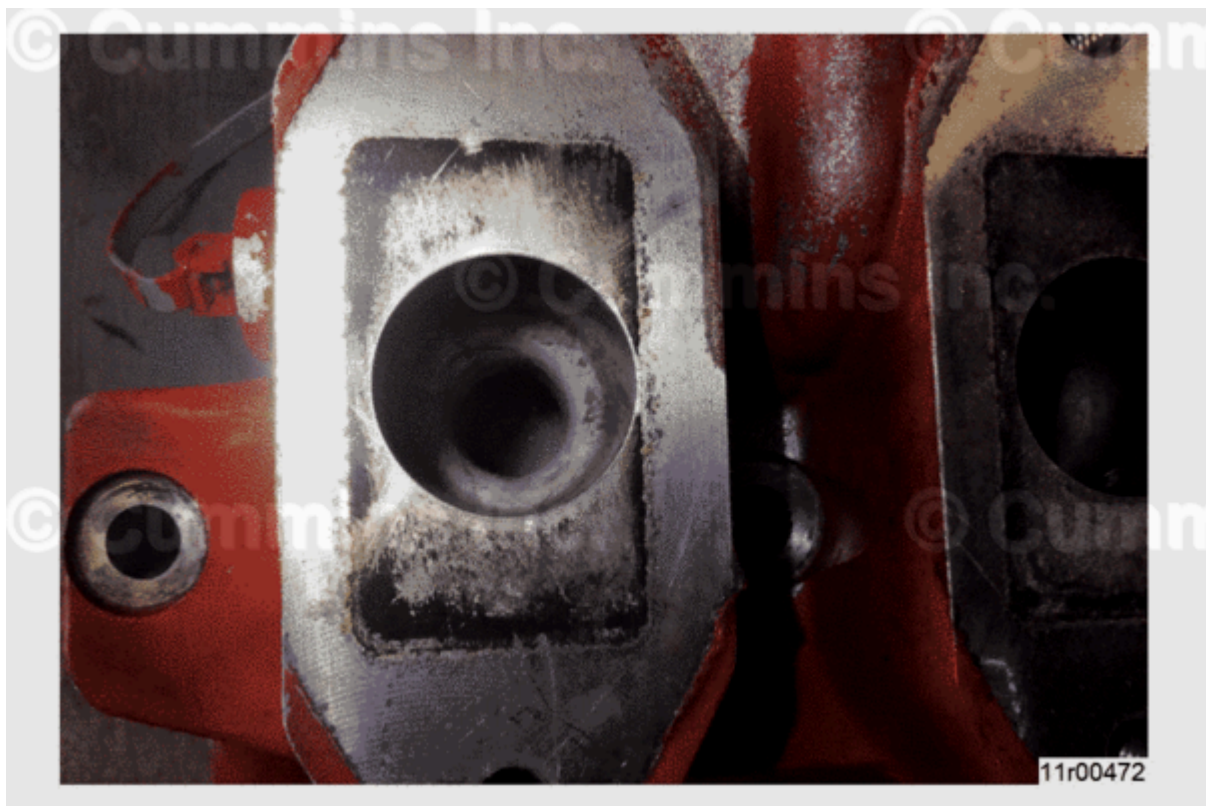


Figure 6, Intake Air Connection after Proper Cleaning.

Table 6, Intake Air Connection after Proper Cleaning

Appearance	Wall inside intake air connection exhaust gas entrance port is visible; exit diameter of exhaust gas entrance port is symmetrical and looks similar compared to a normal intake air connection.
Cause	Intake air connection exhaust gas entrance ports were properly cleaned during repair.
Action	Reuse part; install part on engine to continue with repair.

IV. Inspection and Cleaning of Air Intake Connection Differential Pressure Sensor Ports/Cross Drillings

- Verify the engine is off and that the EGR system has been given time to cool down.
- Remove the Air Intake Connection from the engine. See corresponding Service Manual. Reference Procedure 010-080 in Section 10.
- Reference Figure 1 for an overview of the Air Intake Connection and to locate the EGR differential pressure sensor ports.
- Inspect the EGR differential pressure sensor ports/cross drillings on the Air Intake Connection for any soot buildup. Use a flashlight and place against each port to inspect. Determine if any visible light can be observed on the other end of each port or if any chunks of soot are present. Reference Figures 7, 8, and 9.

- If soot deposits are observed inside the Air Intake Connection EGR differential pressure sensor ports/cross drillings, proceed to thoroughly clean them. Use a nylon brush and solvent to remove all soot found inside. Reference Figures 10-12 for locations of EGR differential pressure sensor ports/cross drillings. See corresponding Service Manual. Reference Procedure 010-080 in Section 10.
- Inspect the EGR differential pressure sensor ports/cross drillings for any visible signs of soot deposits. Use a flashlight to look inside ports and inspect other side of each port to confirm that some light is visible. Reference Figures 13-15 for a visual example.
- Use Figure 1 above to locate the exhaust gas entrance ports. If EGR exhaust gas entrance ports are found to contain excessive soot deposits, reference Section III in this bulletin for instructions on how to clean both sides.
- After cleaning is completed, install the Air Intake Connection onto the engine. See corresponding Service Manual. Reference Procedure 010-080 in Section 10.
- After the Air Intake Connection has been installed on the engine check for leaks on the intake and the exhaust systems. See corresponding Service Manual. Reference Procedure 010-024 in Section 10.



Figure 7, EGR Differential Pressure Sensor Ports of an Air Intake Connection With Excessive Soot.

Table 7, EGR Differential Pressure Sensor Port of an Air Intake Connection With Excessive Soot

Appearance	Soot buildup/plugging observed in the EGR differential pressure sensor port. Light is not observed on the other end of port when using a flashlight.
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Cause	Soot accumulation inside the EGR differential pressure sensor port.
Action	Clean the EGR differential pressure sensor ports with a nylon brush or similar device.



Figure 8, Exhaust Gas Entrance Port of an Air Intake Connection With Excessive Soot.

Table 8, Exhaust Gas Entrance Port of an Air Intake Connection With Excessive Soot	
Appearance	Soot layer observed inside the exhaust gas entrance port; soot buildup/plugging observed at the EGR differential pressure cross drilling. Port/cross drilling is observed to be barely visible or completely missing.
Cause	Soot accumulation taking place inside Air Intake Connection exhaust gas entrance ports and plugging EGR differential pressure cross drilling.
Action	Clean the EGR differential pressure cross drilling and the exhaust gas entrance ports using a nylon brush and solvent.



Figure 9, EGR Differential Pressure Cross Drilling of an Air Intake Connection With Excessive Soot.

Table 9, EGR Differential Pressure Cross Drilling of an Air Intake Connection With Excessive Soot

Appearance	Soot buildup/plugging observed inside the EGR differential pressure cross drilling after removing pipe plug.
Cause	Soot accumulation inside the Air Intake Connection EGR differential pressure cross drilling.
Action	Clean the EGR differential pressure cross drilling with a nylon brush or similar device. If the cross drilling is blocked, it may be necessary to soak the air intake connection in safety solvent before using device to remove the blockage. After cleaning the cross drilling, install the pipe plug. Use thread sealant, Part Number 3375066, or equivalent, to prevent leaks.

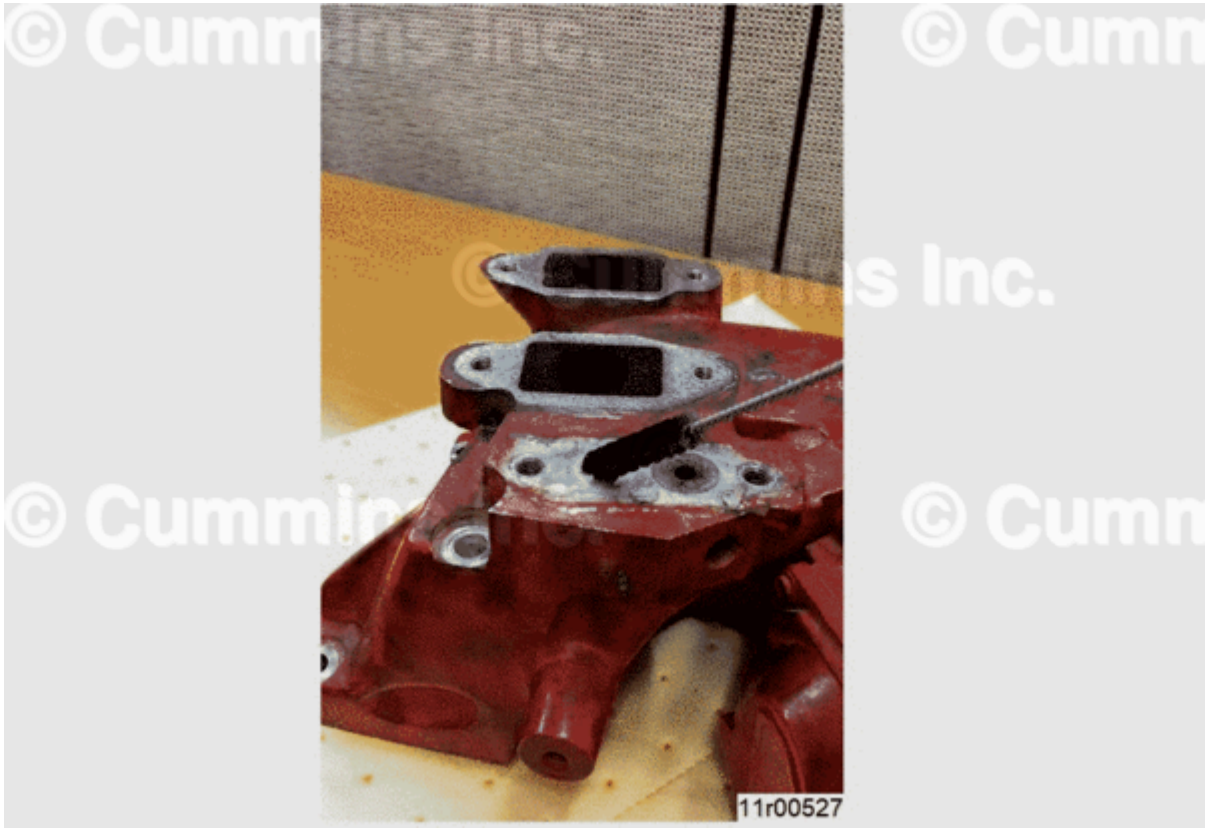


Figure 10, Cleaning EGR Differential Pressure Sensor Port of an Air Intake Connection With Excessive Soot.

Table 10, Cleaning EGR Differential Pressure Sensor Port of an Air Intake Connection With Excessive Soot

Appearance	Nylon brush or similar device and solvent being used to clean soot off inside of EGR differential pressure sensor port.
Cause	-
Action	Complete cleaning of EGR differential pressure sensor port.



Figure 11, Cleaning EGR Differential Pressure Sensor Port of an Air Intake Connection With Excessive Soot.

Table 11, Cleaning EGR Differential Pressure Sensor Port of an Air Intake Connection With Excessive Soot

Appearance	Nylon brush or similar device and solvent being used to clean soot off inside of EGR differential pressure sensor port.
Cause	-
Action	Complete cleaning of EGR differential pressure sensor port.



Figure 12, Cleaning EGR Differential Pressure Cross Drilling of an Air Intake Connection With Excessive Soot.

Table 12, Cleaning EGR Differential Pressure Cross Drilling of an Air Intake Connection With Excessive Soot

Appearance	Nylon brush or similar device and solvent being used to clean soot off inside of EGR differential pressure cross drilling.
Cause	-
Action	Clean the EGR differential pressure cross drilling with a nylon brush or similar device. If the cross drilling is blocked, it may be necessary to soak the air intake connection in safety solvent and use device to remove the blockage. After cleaning the cross drilling, install the pipe plug. Use thread sealant, Part Number 3375066, or equivalent, to prevent leaks.

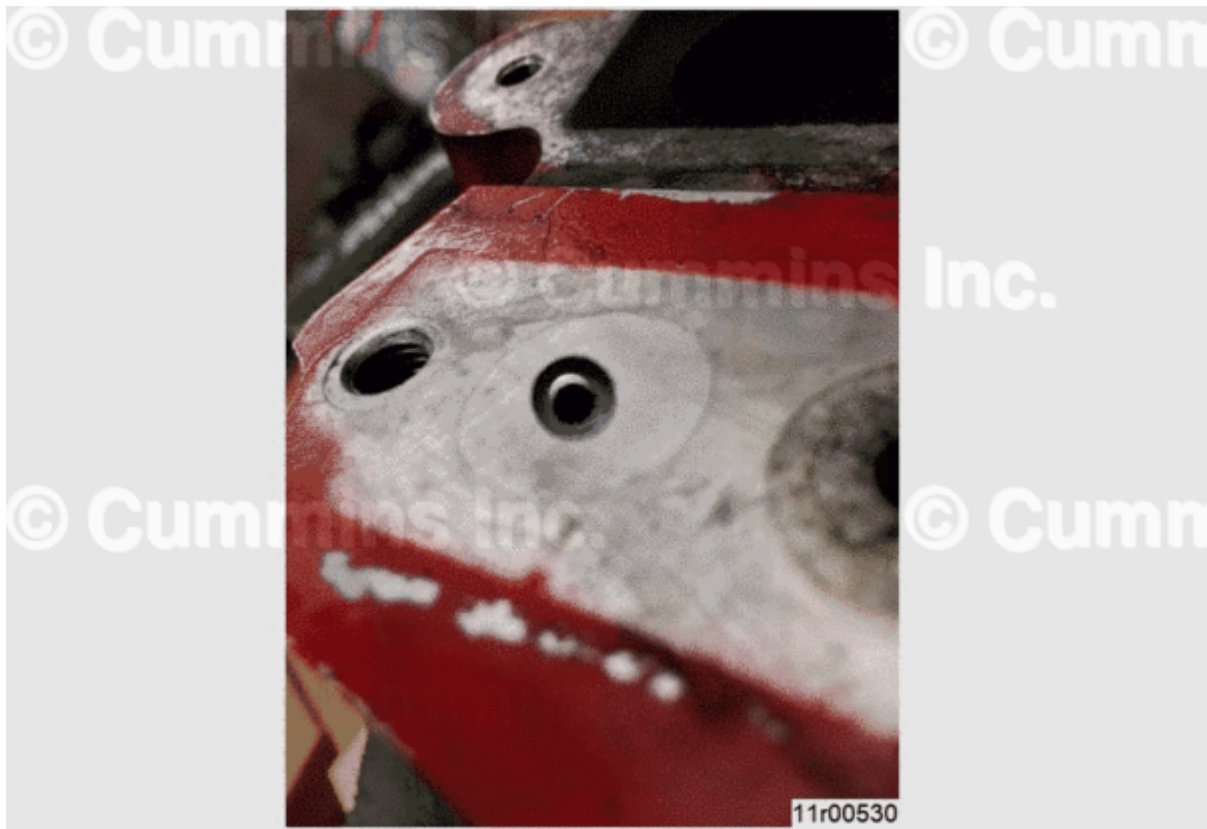


Figure 13, EGR Differential Pressure Sensor Port of an Air Intake Connection After Proper Cleaning.

Table 13, EGR Differential Pressure Sensor Port of an Air Intake Connection After Proper Cleaning

Appearance	Soot buildup is properly cleaned, no port blockage or plugging is observed; diameter of the EGR differential pressure sensor port is symmetrical and flashlight light can be seen.
Cause	EGR differential pressure sensor ports were properly cleaned during repair.
Action	Reuse part; install part on engine to continue with repair.

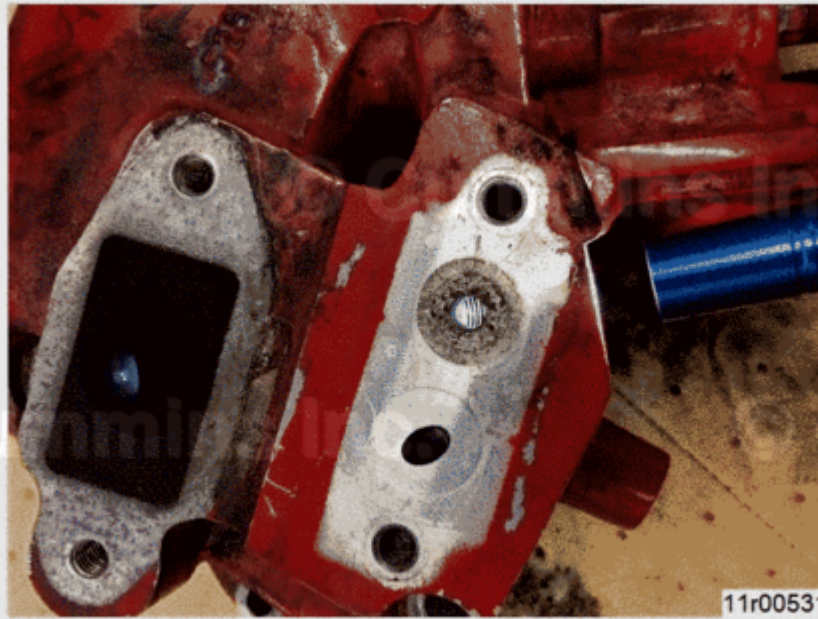


Figure 14, EGR Differential Pressure Sensor Port of an Air Intake Connection After Proper Cleaning.

Table 14, EGR Differential Pressure Sensor Port of an Air Intake Connection After Proper Cleaning

Appearance	Soot buildup is properly cleaned, no port blockage or plugging is observed; diameter of the EGR differential pressure sensor port is symmetrical and flashlight light can be seen.
Cause	EGR differential pressure sensor ports were properly cleaned during repair.
Action	Reuse part; install part on engine to continue with repair.



Figure 15, EGR Differential Pressure Cross Drilling of an Air Intake Connection After Proper Cleaning.

Table 15, EGR Differential Pressure Cross Drilling of an Air Intake Connection After Proper Cleaning

Appearance	Soot buildup is properly cleaned, no blockage or plugging is observed; diameter of the EGR differential pressure cross drilling is symmetrical and flashlight light can be seen.
Cause	EGR differential pressure cross drillings were properly cleaned during repair.
Action	Install the pipe plug. Use thread sealant, Part Number 3375066, or equivalent, to prevent leaks.

Document History

Date	Details
2019-7-22	Module Created
2020-4-13	none
2025-8-19	Fixed typo.

